

Tax Evasion and Productivity

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Recap

Last meeting:

- ▶ Changes in tax evasion over time in response to fiscal policy changes

Feedback:

- ▶ Improve explanation of the empirical strategy
- ▶ In addition to differences over time, also show levels. Maybe levels tell the story better

Today:

- ▶ Add detailed empirical strategy explanation
- ▶ Show levels and differences over time (I think differences over time show a clearer story)
- ▶ Tax evasion heterogeneity across firms with different characteristics, size in particular



Introduction

- ▶ What's the question?
- ▶ Why should you care?

Contribution

- ▶ What do I do?
- ▶ How do I do it?

Overview of Results

- ▶ What do I find?
- ▶ Why is it important?

Model

Empirical Strategies

- ▶ Testing for the presence of tax evasion by industry
 - ▶ Robust to different specifications
- ▶ Regression
 - ▶ Useful to study tax evasion heterogeneity across firms and over time
- ▶ Deconvolution
 - ▶ Recover distribution of tax evasion and productivity
 - ▶ Estimation of unbiased Production Function

Data

Colombia

Testing

Tax Evasion Over Time

Incentives to evade taxes by cost overreporting changed over time due to changes in fiscal policy

The reform of 1983:

- ▶ Differences *across* industries:
 - ▶ Sales Tax (ST) *increased* for most industries from 6% to 10%.
 - ▶ ST-exempt industries *remained exempt* after the reform.
- ▶ Differences *within* industries *across* juridical organizations (JO):
 - ▶ Corporate Income Tax (CIT) increased 8% on average for most Proprietorships (through *Individual Income Tax*), while
 - ▶ CIT decreased 2 percentage points for Limited Liability Companies (LLCs), from 20% to 18%

Empirical Approach

Taking the *FOC* and stacking over industries. For Corporations, the truth-reporters, i in industry j at year t , we have,

$$s_{ijt}^C = \gamma_j + \varepsilon_{ijt} \quad (1)$$

Where s_{ijt}^C is the share of materials' cost over sales, γ_j is the output elasticity of materials for industry j (CD-PF), and ε_{ijt} is the error term

Unincorporated Firms

For the rest of firms, the ST and CIT generated different incentives to evade taxes depending on whether they belong to an ST-exempt industry or not.

For unincorporated firms,

$$s_{ijt}^{NC} = \gamma_j + e_{ijt}^{ST} + e_{ijt}^{CIT} + \varepsilon_{ijt} \quad j \in \text{ST-Liable} \quad (2)$$

$$s_{ijt}^{NC} = \gamma_j + e_{ijt}^{CIT} + \varepsilon_{ijt} \quad j \in \text{ST-Exempt} \quad (3)$$

where e_{ijt}^{ST} and e_{ijt}^{CIT} stand for the overreporting due to sales tax and corporate income tax, respectively.

Combining the Three Types of Firms

- ▶ Let D_{NC} be a dummy indicating whether the firm is not incorporated;
- ▶ D_{NE} , a dummy indicating whether the industry is ST-liable (not exempted of ST); and
- ▶ D_E , a dummy indicating whether the industry is ST-exempted, such that $D_E + D_{NE} = 1$.

Corporations and Unincorporated Firms in ST-Liable Industries

Combining Equation 1 with Equation 2,

$$\begin{aligned}(1 - D_{NC})s_{ijt} + D_{NC}s_{ijt} &= (1 - D_{NC})(\gamma_j + \varepsilon_{ijt}) \\ &\quad + D_{NC}(\gamma_j + e_{ijt}^{ST} + e_{ijt}^{CIT} + \varepsilon_{ijt}) \\ s_{ijt} &= \gamma_j + D_{NC}(e_{ijt}^{ST} + e_{ijt}^{CIT}) + \varepsilon_{ijt} \\ s_{ijt} &= \gamma_j + D_{NC}(e_{ijt}^{ST} + e_{ijt}^{CIT}) + \varepsilon_{ijt} \quad j \in \text{ST-Liable}\end{aligned}$$

Corporations and Unincorporated Firms in ST-Exempt Industries

Likewise, combining Equation 1 with Equation 3,

$$(1 - D_{NC})s_{ijt} + D_{NC}s_{ij't} = \gamma_j + D_{NC}e_{ijt}^{CIT} + \varepsilon_{ijt}$$
$$s_{ijt} = \gamma_j + D_{NC}e_{ijt}^{CIT} + \varepsilon_{ijt} \quad j \in \text{ST-Exempt}$$

All Firms

Combining the firms in both types of industries, we have,

$$\begin{aligned}D_E s_{ijt} + D_{NE} s_{ijt} &= D_E \left(\gamma_j + D_{NC} e_{ijt}^{CIT} + \varepsilon_{ijt} \right) \\&\quad + D_{NE} \left(\gamma_j + D_{NC} (e_{ijt}^{ST} + e_{ijt}^{CIT}) + \varepsilon_{ijt} \right) \\(D_E + D_{NE}) s_{ijt} &= (D_E + D_{NE}) \gamma_j + D_E D_{NC} e_{ijt}^{CIT} \\&\quad + D_{NE} D_{NC} (e_{ijt}^{ST} + e_{ijt}^{CIT}) + (D_E + D_{NE}) \varepsilon_{ijt} \\s_{ijt} &= \gamma_j + D_E D_{NC} e_{ijt}^{CIT} + D_{NE} D_{NC} (e_{ijt}^{ST} + e_{ijt}^{CIT}) + \varepsilon_{ijt}\end{aligned}\tag{4}$$

Taking Expectations

$$E[s_{ijt}|D_{NC} = 0] = \gamma_j$$

$$E[s_{ijt}|D_{NC} = 1, D_E = 1, D_{NE} = 0] - E[s_{ijt}|D_{NC} = 0] = \mu_E^{CIT}$$

$$E[s_{ijt}|D_{NC} = 1, D_E = 0, D_{NE} = 1] - E[s_{ijt}|D_{NC} = 0] = \mu_{NE}^{ST+CIT}$$

we can learn the average overreporting by firms in ST-exempt, μ_E^{CIT} , and ST-liable industries, μ_{NE}^{ST+CIT}

Changes Over Time

In the case of the fiscal policy changes, we are interested in the changes in average tax evasion after the 1983 fiscal reform.

We can take the expectation difference, before t and after the reform t'

$$E[s_{ijt}|D_{NC} = 1, D_E = 1, D_{NE} = 0, t'] \\ - E[s_{ijt}|D_{NC} = 1, D_E = 1, D_{NE} = 0, t] = \Delta_E^{CIT}$$

$$E[s_{ijt}|D_{NC} = 1, D_E = 0, D_{NE} = 1, t'] \\ - E[s_{ijt}|D_{NC} = 1, D_E = 0, D_{NE} = 1, t] = \Delta_{NE}^{CIT+ST}$$

Where $\Delta_E^{CIT} = \mu_{E,t'}^{CIT} - \mu_{E,t}^{CIT}$ and $\Delta_{NE}^{CIT+ST} = \mu_{NE,t'}^{CIT+ST} - \mu_{NE,t}^{CIT+ST}$

Expected Changes in Tax Evasion After the Fiscal Reform of 1983

- ▶ Since most ST-liable industries received an increase in the tax rate, we expect an increase in average tax evasion by these firms.
- ▶ With respect to CIT, the expected change is ambiguous.
 - ▶ Proprietorships faced an increase in their incentives to evade taxes, while
 - ▶ LLCs a small decrease.
 - ▶ The net effect depends on the relative weight of each JO and the magnitude of the change in their respective CIT.

In Regression Form

Equation 4 becomes,

$$s_{ijt} = D_{NC} D_E \mathbf{D}_t \Gamma'_{0,t} + D_{NC} D_{NE} \mathbf{D}_t \Gamma'_{1,t} + \gamma_j + \varepsilon_{ijt} \quad (5)$$

- ▶ Where $\mathbf{D}_t$ is a $1 \times T$ vector of year dummies, and
- ▶ $\Gamma_{0,t}$ and $\Gamma_{1,t}$ are the $1 \times T$ vectors of coefficients
 - ▶ $\Gamma_{0,t}$: average tax evasion of firms in ST-exempt industries due to CIT
 - ▶ $\Gamma_{1,t}$: average tax evasion of firms in ST-liaible industries due to ST and CIT
- ▶ γ_j are now industry fixed effects.
- ▶ D_{NC} , D_E , D_{NE} , and ε_{ijt} are defined as before.

Differences Over Time in Regression Form

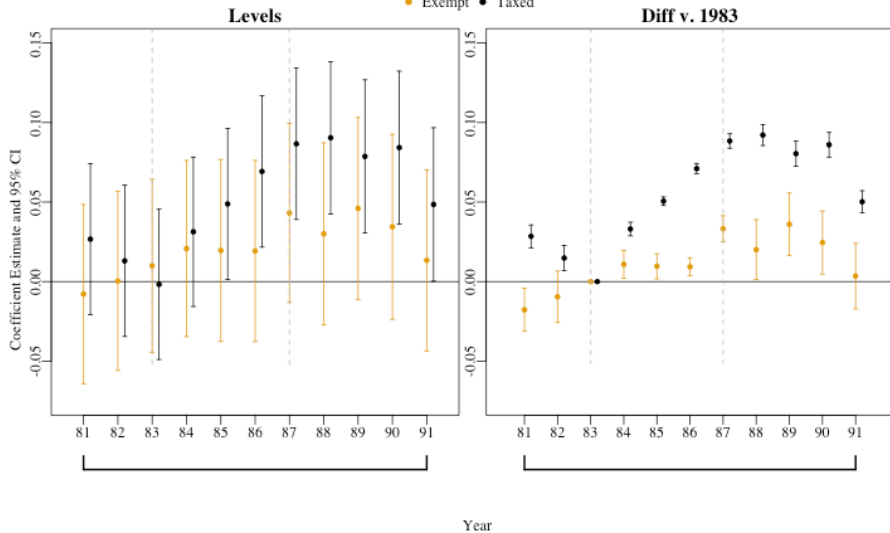
$$\begin{aligned}s_{ijt} = & \beta_0 D_{NC} + \beta_1 D_{NC} D_E + \beta_2 D_{NC} D_{NE} \\ & + D_{NC} D_E \tilde{\mathbf{D}}_t \tilde{\Gamma}'_{0,t} \\ & + D_{NC} D_{NE} \tilde{\mathbf{D}}_t \tilde{\Gamma}'_{1,t} + \gamma_j + \varepsilon_{ijt}\end{aligned}\tag{6}$$

- ▶ Where now $\tilde{\mathbf{D}}_t$ is a $1 \times (T - 1)$ vector of year dummies excluding 1983, and
- ▶ $\tilde{\Gamma}_{0,t}$ and $\tilde{\Gamma}_{1,t}$ are the corresponding vectors of coefficients
 - ▶ $\tilde{\Gamma}_{0,t}$: change in average tax evasion due to changes in CIT for firms in ST-exempt industries relative to 1983
 - ▶ $\tilde{\Gamma}_{1,t}$: change in average tax evasion due to changes in ST and CIT for firms in ST-liable industries relative to 1983.

Tax Evasion by Unincorporated Firms

Input Overreporting in ST-Exempt and ST-Liable Industries

● Exempt ● Taxed



Unincorporated Firms in Both Types of Industries

After the fiscal reform of 1983, we observe

- ▶ an increase in tax evasion for unincorporated firms in both types of industries,
- ▶ and the increase is larger for firms facing CIT and ST-liable industries.
- ▶ In ST-exempt industries, the increase implies that the effect of the increase in CIT for Proprietorships dominates the effect of the decrease in CIT for LLCs.

Why Do The Standard Errors Look Change So Much?

My guess:

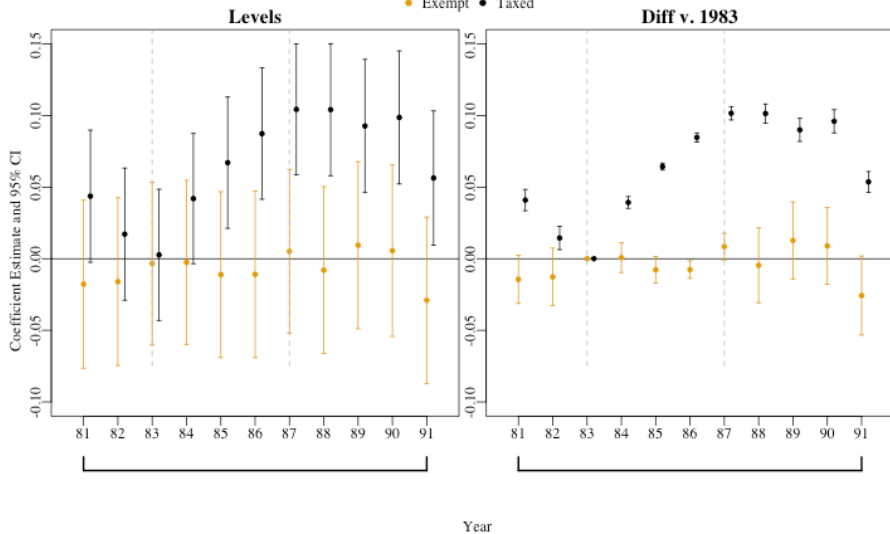
- ▶ Levels: Difference between Corporations and Unincorporated firms. Unincorporated firms are more heterogeneous in terms of tax evasion, and thus the standard errors are larger.
- ▶ Differences over time: Difference overtime but with themselves. Since we are comparing the same type of firms over time, we are capturing the reaction to the fiscal policy change, and this reaction is more homogeneous across firms, thus the standard errors are smaller.
 - ▶ The firm that evaded more before the reform evades more after the reform, and the firm that evaded less before the reform evades less after the reform, but the increase in evasion is similar across firms

Splitting Unincorporated Firms by JO into LLCs and Proprietorships

Tax Evasion by LLCs

Input Overreporting in ST-Exempt and ST-Liable Industries

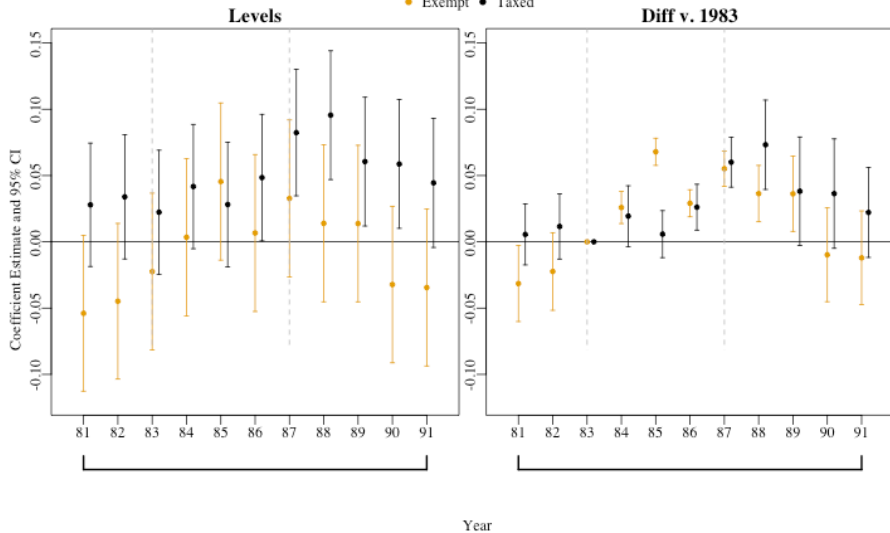
● Exempt ● Taxed



Tax Evasion by Proprietorships

Input Overreporting in ST-Exempt and ST-Liable Industries

● Exempt ● Taxed



LLCs vs Proprietorships in Both Types of Industries

- ▶ LLCs in ST-liable industries show a significant increase in tax evasion after the fiscal reform of 1983, while LLCs in ST-exempt industries show no significant change.
- ▶ Proprietorships in ST-exempt industries show a moderate but significant increase in tax evasion after the fiscal reform of 1983
- ▶ Proprietorships in ST-liable industries show no immediate change in tax evasion after the fiscal reform of 1983
 - ▶ Proprietorships show a small but significant increase in tax evasion starting in 1986 and until 1989. But this increase might be due to the 1986 reform.

Takeaways

- ▶ As expected, firms are adjusting their tax evasion by overreporting their inputs (materials) in response to changes in both CIT and ST (VAT).
- ▶ Overall, firms with larger increases in their incentives to evade taxes, such as firms facing large increases in tax rates after the fiscal reform of 1983, show larger increases in tax evasion.
- ▶ Firms with a decrease in their incentives to evade taxes after the fiscal reform of 1983, such as LLCs in ST-exempt industries, show no significant change in tax evasion, suggesting firms might respond *asymmetrically* to changes in their incentives to evade taxes.

Tax Evasion and Firm Heterogeneity

- ▶ Are there systematic differences in tax evasion across firms with different characteristics?
- ▶ Why? Firms might face different probabilities of detection, cost of evading taxes, and even different tax rates —thus, different incentives— depending on their characteristics.
- ▶ Size for example.
 - ▶ Common intuition is that larger firms face higher probabilities of detection, and thus evade less.
 - ▶ However, recent evidence suggests that tax evasion increases along the size distribution (revenue) and then drops at the top of the distribution (Carrillo et al. 2022).

Empirical Strategy

Same as before, for the truth-reporters, corporations, we have,

$$s_{ijt}^C = \gamma_j + \varepsilon_{ijt}$$

For the evaders, non-corporates, we now have,

$$s_{ijt}^{NC} = \gamma_j + e_{ijt}(\tau, q, \kappa) + \varepsilon_{ijt} \quad (7)$$

- ▶ where the tax evasion, $e_{ijt}(\tau, q, \kappa)$, is a function of
- ▶ the effective sales tax rate τ ,
- ▶ the probability of detection q and
- ▶ the cost of evasion κ .
- ▶ γ_j and ε_{ijt} are defined as before.

Empirical Strategy (Cont.)

Combining the two types of firms, we have,

$$\begin{aligned}(1 - D_{NC})s_{ijt} + D_{NC}s_{ijt} &= (1 - D_{NC})(\gamma_j + \varepsilon_{ijt}) \\ &\quad + D_{NC}(\gamma_j + e_{ijt}(q, \kappa) + \varepsilon_{ijt}) \quad (8) \\ s_{ijt} &= \gamma_j + D_{NC}e_{ijt}(\tau, q, \kappa) + \varepsilon_{ijt}\end{aligned}$$

► where D_{NC} is defined as before

Empirical Strategy (Cont.)

- ▶ Since tax evasion $e_{ijt}(\cdot)$, the probability of detection q , and the cost of evasion κ are all **unobserved**, we could use **firm characteristics** that are likely to influence them
- ▶ In theory, error terms is independent of most firm characteristic that could influence tax evasion $\mathbb{E}[\varepsilon_{ijt}|e_{ijt}] = 0$.
- ▶ In practice however I would like to avoid variables that could be **mechanically correlated** with the explained variable, the input cost share.
 - ▶ For example, current size measured as revenue could be mechanically correlated with the denominator of the input cost share.

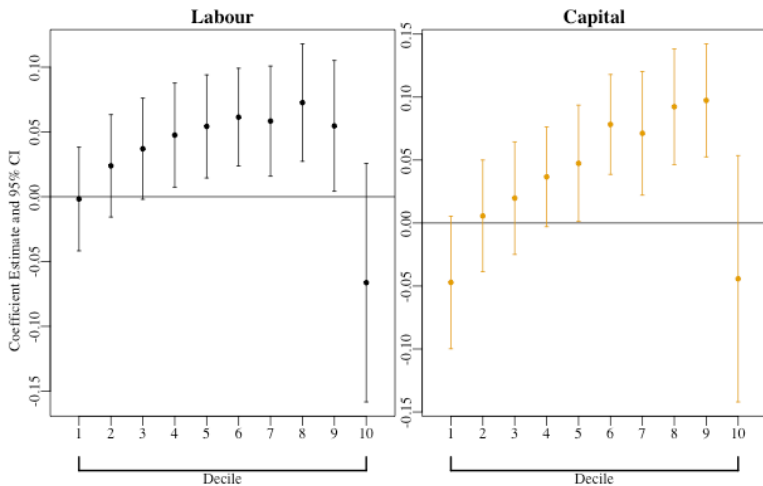
Regression Specification

To estimate Equation 8, I use the following regression specification,

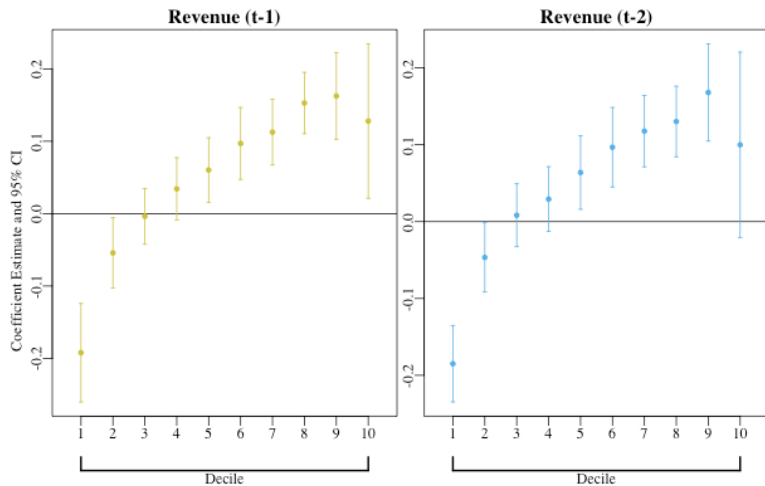
$$s_{ijt} = \gamma_j + D_{NC} \mathbf{X}_{it} \Gamma'_0 + \varepsilon_{ijt} \quad (9)$$

- ▶ $\mathbf{X}_{it}$ is a $1 \times K$ vector of dummy variables for each decile of size measured by labor, capital, last-year revenue, and two-years-ago revenue
- ▶ Γ_0 is a $1 \times K$ vector of coefficients which captures the average tax evasion by firms in each decile of size

Average Tax Evasion by Size Decile



Average Tax Evasion by Size Decile



Tax Evasion and Size

- ▶ Similar pattern across size measures: Tax evasion increases along the size distribution and then drops at the top of the distribution
- ▶ Surprisingly consistent with recent evidence in the literature (Figure 1).
- ▶ Large firms evade more than small firms, but the largest firms do not evade taxes.

But Why?

- ▶ Since the probability of detection cannot be, by logic, decreasing in size, it should at least be non-decreasing.
 - ▶ Most likely it is flat along the size distribution, and then increases rapidly at the top of the distribution.
- ▶ Then, it follows that the cost of evasion should be decreasing along the size distribution, and then it might (or might not) increase at the top of the distribution.

Marginal Effects

- ▶ All previous measures were correlated with size, is there one more relevant than the others?
- ▶ What is the marginal effect of each size measure holding the others constant?

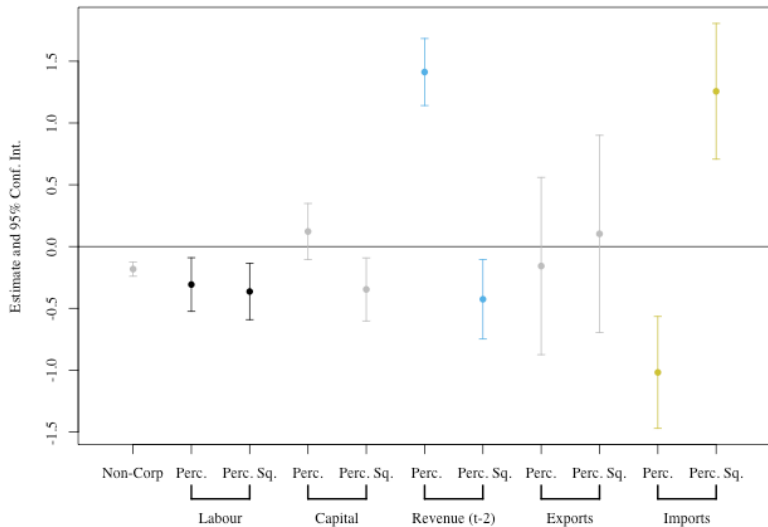
I use the following regression specification of Equation 8 to investigate these questions,

$$s_{ijt} = \gamma_j + \beta_0 D_{NC} + D_{NC} \mathbf{X}_{it} \Gamma'_0 \\ + D_{NC} \mathbf{X}_{it}^2 \Gamma'_1 + \varepsilon_{ijt}$$

- ▶ where now $\mathbf{X}_{it}$ stands for the **percentiles** of: labour, capital, revenue (lagged two years), exports, and imports.
- ▶ The square term is included to capture the non-linearities of the previous estimation.

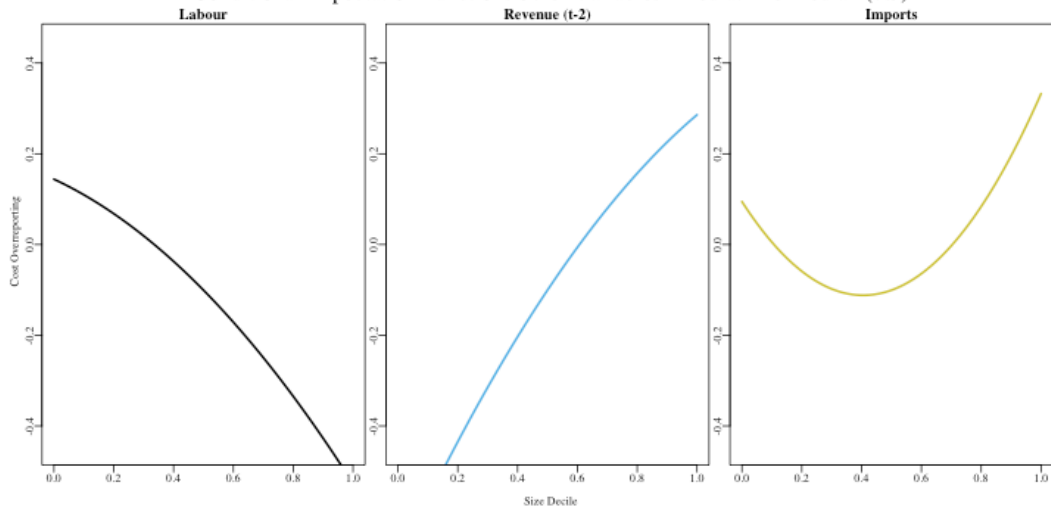
Tax Evasion and Firm Size

Conditional Effects of Size on Tax Evasion



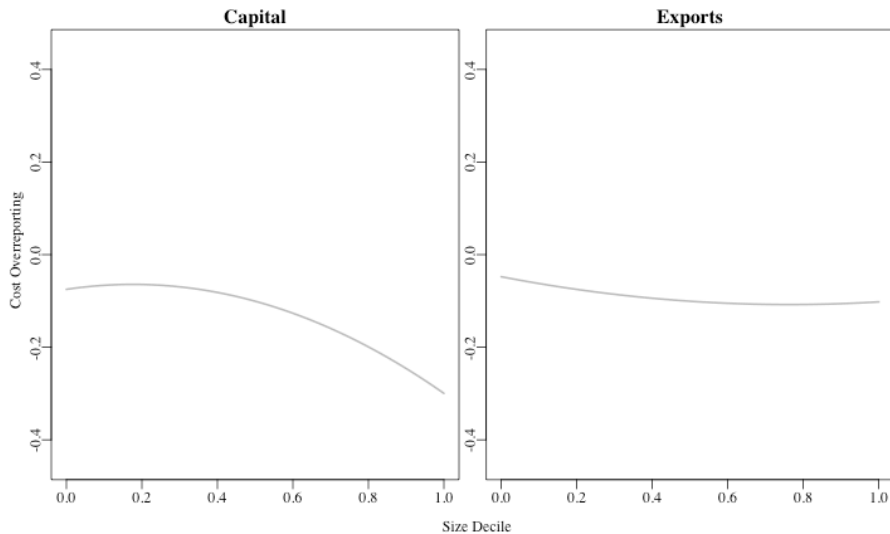
Expected Tax Evasion

Conditional Expectation Function. Other Variables Fixed at The Median (0.5)



Expected Tax Evasion

Conditional Expectation Function. Other Variables Fixed at The Median (0.5)



Marginal Effects of Different Size Measures

Holding other size measures constant, tax evasion

- ▶ decreases at a marginal increasing rate (decreasing and concave) as firms grow in labour
- ▶ increases at a marginal decreasing rate (increasing and concave) as firms grow in revenue ($t-2$)
- ▶ decreases at a marginal decreasing rate (decreasing and convex) as firms grow in imports

Does This Make Sense? Labour

Comparing two firms with similar characteristics but different percentiles of labour, the firm with a *higher* percentile of labour *evades less*

- ▶ As firms grow in labour, they face *higher probabilities of detection*, as the probability of being denounced to the tax authority by an employee increases
- ▶ They might also face *higher costs of evading taxes* as more employees need to be involved in the evasion scheme (keeping double accounting books, larger accounting department, warehouse employees might also need to be involved, etc.)

Revenue

Comparing two firms with similar characteristics but different percentiles of revenue (t-2), the firm with a higher percentile of revenue evades more

- ▶ As firms grow in revenue, they have more resources to allocate to tax evasion, and thus face *lower costs* of evading taxes relatively to low revenue firms
- ▶ The *cost of evasion* might *abruptly increase for the largest* firms.
 - ▶ The largest firms might be able to implement more sophisticated but *legal tax avoidance* strategies (e.g. profit offshoring), and thus the cost of evading taxes *illegally* increases significantly relative to legal schemes.
 - ▶ The reputational cost of being caught evading taxes might be higher for the largest firms, as they are more visible to the public and potential investors.
 - ▶ *If I get caught evading, I'm a criminal, but if I get caught avoiding taxes, I'm just a smart businessman.*

Revenue (Cont.)

- ▶ Higher revenue might also increase the probability of detection, as visibility increases. The evidence suggests that this increase in the detection probability should be relatively smaller than the decrease in the evasion cost for most of the size distribution
- ▶ However, at the top of the revenue distribution, the probability of detection might increase significantly as it becomes strategically targeted by the tax authority, and thus tax evasion drops for the largest firms

Imports

- ▶ Imports in Colombia are subject to ST on top of the regular import duties.
- ▶ Imports are also harder to misreport because they leave larger paper trails, and thus the authority can more easily trace them.
- ▶ Consequently, a larger share of imports increase tax evasion incentives
 - ▶ larger share of imports increases the share of total material costs that cannot be claimed as deductions from ST, resulting in lower net after-tax profits, and thus increasing incentives for tax evasion

Imports (Cont.)

This explains why firm in the highest percentiles of imports evade more

- ▶ At lower levels (below the 40th perc.), incentives are small. Evasion looks like it decreases with imports at low levels
- ▶ as firms move above the 40th percentile however, the incentives to evade taxes increase significantly

Takeaways

Overall, these results suggest that the non-linear pattern observed in the previous section might be explained by:

1. a probability of detection that remains flat (or increases slowly) along most of the size distribution, but then increases sharply for the largest firms.
 - ▶ The detection probability increases sharply for the largest firms because they are more visible to the tax authority, and because
 - ▶ these firms are more likely to be denounced to the tax authority by an employee

Takeaways (Cont.)

2. a cost of evasion that decreases along the size distribution (but then it might increase rapidly for the largest firms). This is because
 - ▶ firms with small revenues face higher relative costs of evasion as they have fewer resources to allocate to tax evasion. As firms increase in revenue, they can allocate more resources to tax evasion, reducing the relative cost of evasion.
 - ▶ For the largest revenue firms however, the cost of evasion might increase abruptly due to the higher costs of illegal evasion relative to legal avoidance strategies accessible mostly for the largest firms only, and higher reputational costs if caught evading taxes.

Appendix

B) Value of Reported Ghost Purchases Over Firm Total Purchases, by Firm Size

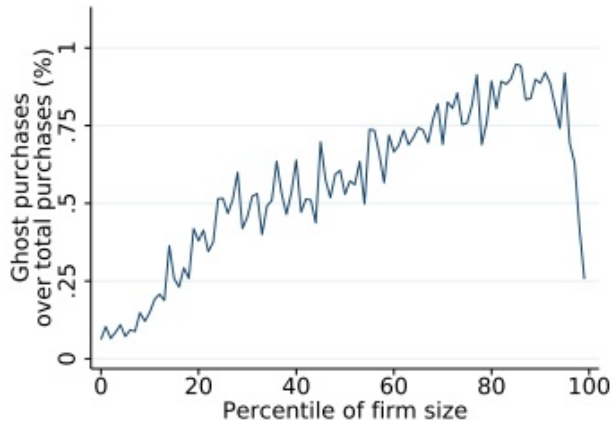
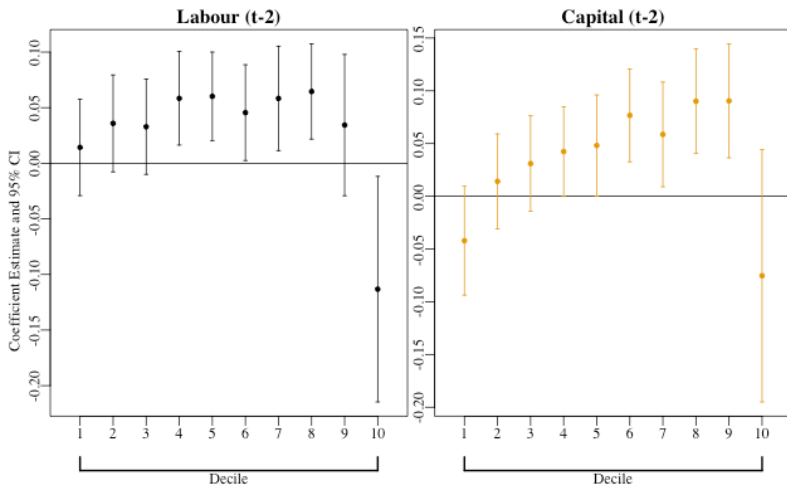


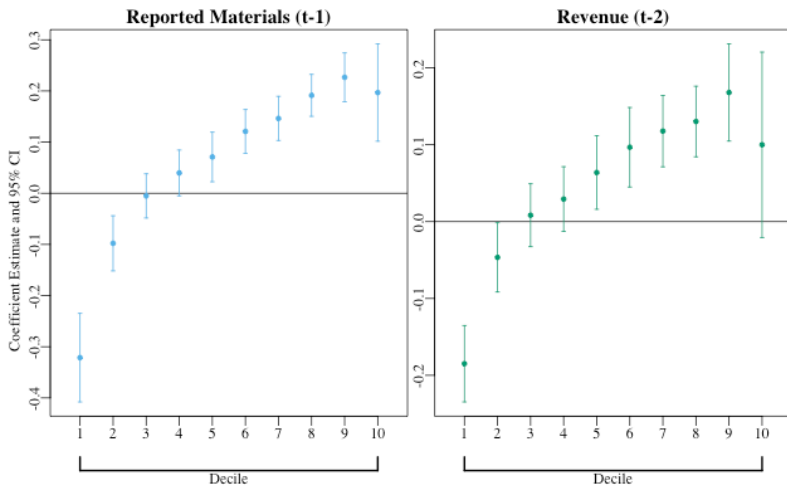
Figure 1: Figure 1B from Carrillo et al. (2022) ←↻

Robust Check for Size Heterogeneity

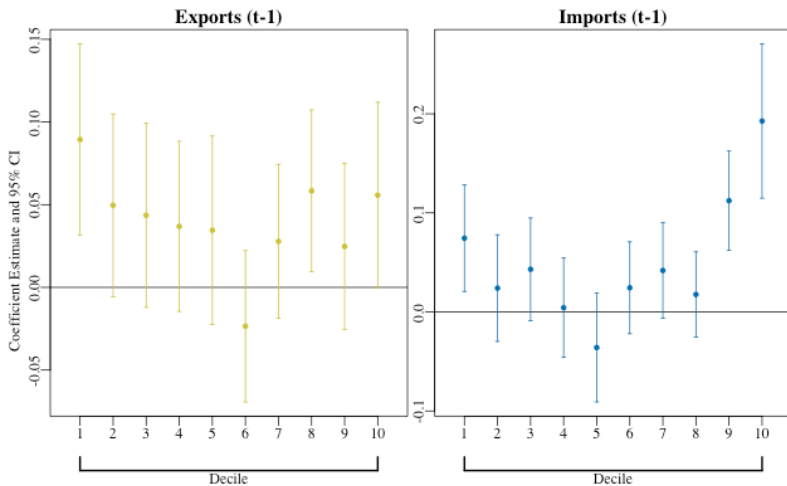
Average Tax Evasion by Size Decile



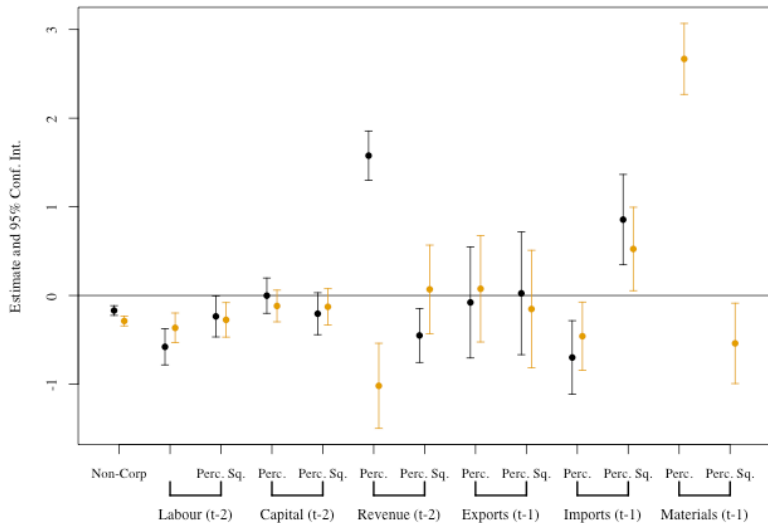
Average Tax Evasion by Size Decile



Average Tax Evasion by Size Decile



Tax Evasion and Size: Marginal Effects



References

Carrillo, Paul, Dave Donaldson, Dina Pomeranz, and Monica Singhal. 2022.
“Ghosting the Tax Authority: Fake Firms and Tax Fraud.” NBER.
<http://www.nber.org/papers/w30242>.